The empirical recurrence for OEIS A234446: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A234446 records "Empirical recurrence of order 92", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 625$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 92, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 92 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234446 is "Number of (n+1) X (3+1) 0..4 arrays with every 2 X 2 subblock having its diagonal sum differing from its antidiagonal sum by 1 (constant-stress 1 X 1 tilings).". It has offset 1 and begins

$$8064, 125080, 2053280, 34601928, 588663528, 10079824144, 172781443608, 2969800840496, \ldots$$

The entry states:

> Empirical recurrence of order 92 (see link above).

As of the "Last modified" line on the live entry (Jun 20 2022 18:37:29, revision 8) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 625$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 625$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1250$ exact terms of the model returns order 92 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{92} c_i\, a(n-i),$$

c_1	0	c_{24}	−7410149520729544915868383 2	c_{47}	
c_2	1512	c_{25}	0	c_{48}	−201228612116073294
c_3	0	c_{26}	206457044271745026049791677 9	c_{49}	
c_4	−998072	c_{27}	0	c_{50}	7421552113174267665
c_5	0	c_{28}	−4798206739541463335117929513 7	c_{51}	
c_6	383138936	c_{29}	0	c_{52}	−2310778000132063919
c_7	0	c_{30}	9346785371696284413893161002 42	c_{53}	
c_8	−96530056587	c_{31}	0	c_{54}	6058337175674623479 4
c_9	0	c_{32}	−153202890668872080937150309353 38	c_{55}	
c_{10}	17124826083632	c_{33}	0	c_{56}	−1333380071938141285
c_{11}	0	c_{34}	211949858139742719231322348759152	c_{57}	
c_{12}	−2241191424532657	c_{35}	0	c_{58}	2454924967763120122 7
c_{13}	0	c_{36}	−2480793835991680776528564383910724	c_{59}	
c_{14}	223620668600416008	c_{37}	0	c_{60}	−3766075838929486655
c_{15}	0	c_{38}	24608494270801441712566697988161272	c_{61}	
c_{16}	−17427288482359214080	c_{39}	0	c_{62}	4792751130060520578 9
c_{17}	0	c_{40}	−207107228511037403128531196171606472	c_{63}	
c_{18}	1080392086333608736685	c_{41}	0	c_{64}	−5034760534397109948
c_{19}	0	c_{42}	1479608311101310282497353508626567376	c_{65}	
c_{20}	−54040789109953781386900	c_{43}	0	c_{66}	4341724049455825623 5
c_{21}	0	c_{44}	−8972655743456782528309440110440483904	c_{67}	
c_{22}	2205500186385719289060810	c_{45}	0	c_{68}	−3054275467335037052
c_{23}	0	c_{46}	46159729761863600988203213985732990976	c_{69}	

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 92, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $92+4$ terms removed returns order 92 as well, so $\deg q_{\min} = 92 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $92+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 92 that this sequence satisfies — and that family turns out to have exactly one member.

References

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